



Computer System and Architecture

Machine Problem

MIDTERM

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List the steps that you perform to come up with the program.

Paste the source code and explain the functionalities and take a screenshot of the final output.

Step 1: Planning What the Program Does (Requirement Analysis)

Before writing any code, we planned the overall flow of the program. The goal was to build a fun terminal Slambook in Assembly language that does three things in order:

- Ask the user 20 questions one by one.
- Show an animated loading bar to simulate processing the data.
- Print out a clean, nicely framed Slambook Card showing all the answers.

Step 2: Setting Up Memory Spaces (Buffer & Data Structuring)

In Assembly, you have to manually tell the computer where to store words and characters:

- Storage for Answers (.bss section): We reserved 64 blank spots in memory for each of the 20 questions so the program has space to store whatever text the user types.
- Fixed Text & Designs (.data section): We typed out all the permanent text ahead of time—like the question prompts, decorative borders, title ASCII art, and progress bar frames.

Step 3: Asking Questions and Saving Answers (Input Handling)

To keep the code clean and avoid repeating instructions 20 times, we created a single reusable recipe (procedure) called `get_input`:

- It displays a prompt (like 1. Full Name:).
- It asks Linux to pause and wait for the user to type an answer and press Enter.
- It safely saves whatever was typed straight into the designated memory spot for that question.

Step 4: Making the Animated Progress Bar (Loading Animation)

To create a loading screen effect:

- The program prints a progress bar step (like [10%], then [30%], up to [100%]).



- Between each step, we tell Linux to "sleep" (pause execution) for a fraction of a second using `sys_nanosleep`. Without this delay, the loading bar would flash on screen and finish instantly before you could even see it move!

Step 5: Fixing the Text Overlap Bug (Dynamic Length Calculation)

Earlier, the program tried to print a fixed size of 64 characters for every answer. This caused a big bug where short answers (like Age: 21) left random garbage characters or pulled text from other lines into the frame.

- The Fix: We created a counter loop (`print_string_null`) that scans each answer byte-by-byte until it hits the end of the text.
- It counts the exact number of letters typed and tells Linux to print only that exact length, fixing the overlapping text completely.

Step 6: Formatting the Final Card (Card Assembly)

Finally, we put everything together inside a visual frame:

- For each row, the program prints the left border (`|`), the field title (like Full Name:), the user's saved answer, and the right border.
- Repeating this row-by-row creates a neatly aligned box around your profile without any messy text spilling over the edges.

Memory Management (.bss vs .data):

- `.data`: Houses immutable ASCII string constants, ASCII art components, and ANSI escape codes.
- `.bss`: Holds uninitialized memory (`resb 64`) allocated for storing user responses dynamically.

`get_input`

- Procedure: Encapsulates reading from `stdin` (`int 0x80, eax = 3`). It receives the prompt text address in `ESI` and the destination buffer address in `EDI`.

`print_string_null`

- Procedure: Scans memory addresses sequentially until encountering a null byte (`0`). It subtracts the base pointer from the current address pointer to compute the



exact byte count required for sys_write (eax = 4), preventing memory bleeding across adjacent rows.

sleep_delay

- Procedure: Calls Linux sys_nanosleep (eax = 162) using the timespec struct to pause terminal updates during the loading screen sequence.

clear_screen

- Procedure: Sends the ANSI control sequence \x1B[2J\x1B[H to erase standard output and reset cursor position to top-left (0,0).

SOURCE CODE:

section .data

; ANSI Screen Clear & Cursor Home

clr_scr db 0x1B, "[2J", 0x1B, "[H", 0

clr_len equ \$ - clr_scr

; Fancy Decorative ASCII Borders

border_top db " .-----", 10, 0

l_btop equ \$ - border_top

border_mid db "|-----|", 10, 0

l_bmid equ \$ - border_mid

border_bot db "'-----'", 10, 0

l_bbot equ \$ - border_bot

side_border db "|", 0

l_side equ \$ - side_border

; Main Header Banner

str_title db 10, " .-----", 10

db "|", 10

db "| / _ \| | / \ | V | _) / _ \| / / |", 10

db "| \ _ \| | / \ | V | _ \| | _ \| ' / |", 10

db "| _) | _ \| | | | | | | | | | \ |", 10

db "| | _ \| / _ \| / \ \| | _ \| / \ () _ \| | \ |", 10

db "'-----'", 10, 10, 0



```
len_title equ $ - str_title

; ASCII Photo Frame
str_flower db " | ^_^ ( ^_^ ) +-----+ | ", 10
            db " | ( o.o ) ( 'x' ) | [o_o] OFFICIAL ASSEMBLY SLAMBOOK | | ", 10
            db " | > ^ < c(\"")(\"") +-----+ | ", 10, 0
len_flower equ $ - str_flower

; Loading Screen Emoticons & Progress Bar
str_load_hdr db 10, " [!] Processing Slambook Memory Buffers... Please Wait! ", 10,
10, 0
len_load_hdr equ $ - str_load_hdr

emo_kirby1 db " c('c) (>^_^)> ", 0
len_emo1 equ $ - emo_kirby1
emo_kirby2 db " <('-'<) ^('-' )^ ", 0
len_emo2 equ $ - emo_kirby2
emo_kirby3 db " (>'-'>) Q('Q) <(-^,^)=b ", 10, 0
len_emo3 equ $ - emo_kirby3

pbar_0 db " [ 0%] ██████████", 13, 0
len_p0 equ $ - pbar_0
pbar_10 db " [ 10%] ██████████", 13, 0
len_p10 equ $ - pbar_10
pbar_30 db " [ 30%] ██████████", 13, 0
len_p30 equ $ - pbar_30
pbar_60 db " [ 60%] ██████████", 13, 0
len_p60 equ $ - pbar_60
pbar_80 db " [ 80%] ██████████", 13, 0
len_p80 equ $ - pbar_80
pbar_100 db " [100%] ██████████", 10, 10, 0
len_p100 equ $ - pbar_100

str_uploaded db " (data) Upload complete! Formatting Slambook Card...", 10, 10, 0
len_uploaded equ $ - str_uploaded

str_out_hdr db 10, " .-----.", 10
            db " | OFFICIAL SLAMBOOK CARD | ", 10
            db " '-----' ", 10, 0
len_out_hdr equ $ - str_out_hdr

; Prompts (20 Items)
p_name db " 📄 [01] Full Name : ", 0
```



p_nick db " [02] Nickname : ", 0
p_gender db " [03] Gender : ", 0
p_age db " [04] Age : ", 0
p_bday db " [05] Birthday : ", 0
p_hair db " [06] Hair Color : ", 0
p_height db " [07] Height (m) : ", 0
p_town db " [08] Hometown : ", 0
p_prog db " [09] Program & School : ", 0
p_work db " [10] Job/Workplace : ", 0
p_lang db " [11] Languages Spoken : ", 0
p_happy db " [12] Happiest Moment : ", 0
p_color db " [13] Favorite Color(s) : ", 0
p_movie db " [14] Favorite Movie : ", 0
p_music db " [15] Favorite Music/Song : ", 0
p_singer db " [16] Favorite Singer(s) : ", 0
p_food db " [17] Favorite Food(s) : ", 0
p_hobby db " [18] Hobbies : ", 0
p_book db " [19] Favorite Book : ", 0
p_motto db " [20] Life Motto : ", 0

; Card Output Field Labels (20 Items)

lbl_name db " Full Name : ", 0
lbl_nick db " Nickname : ", 0
lbl_gender db " Gender : ", 0
lbl_age db " Age : ", 0
lbl_bday db " Birthday : ", 0
lbl_hair db " Hair Color : ", 0
lbl_height db " Height : ", 0
lbl_town db " Hometown : ", 0
lbl_prog db " Program/School : ", 0
lbl_work db " Job/Workplace : ", 0
lbl_lang db " Languages : ", 0
lbl_happy db " Happiest Moment: ", 0
lbl_color db " Fav Color(s) : ", 0
lbl_movie db " Fav Movie : ", 0
lbl_music db " Fav Music/Song : ", 0
lbl_singer db " Fav Singer(s) : ", 0
lbl_food db " Fav Food(s) : ", 0
lbl_hobby db " Hobbies : ", 0
lbl_book db " Fav Book : ", 0



```
lbl_motto db " Life Motto  : ", 0
```

```
timespec:
```

```
tv_sec dd 0
```

```
tv_nsec dd 180000000
```

```
section .bss
```

```
buf_name resb 64
```

```
buf_nick resb 64
```

```
buf_gender resb 64
```

```
buf_age resb 64
```

```
buf_bday resb 64
```

```
buf_hair resb 64
```

```
buf_height resb 64
```

```
buf_town resb 64
```

```
buf_prog resb 64
```

```
buf_work resb 64
```

```
buf_lang resb 64
```

```
buf_happy resb 64
```

```
buf_color resb 64
```

```
buf_movie resb 64
```

```
buf_music resb 64
```

```
buf_singer resb 64
```

```
buf_food resb 64
```

```
buf_hobby resb 64
```

```
buf_book resb 64
```

```
buf_motto resb 64
```

```
section .text
```

```
global _start
```

```
_start:
```

```
; 1. Title Banner
```

```
call clear_screen
```

```
mov ecx, str_title
```

```
mov edx, len_title
```

```
call print_string_len
```

```
mov ecx, border_top
```

```
mov edx, l_btop
```

```
call print_string_len
```



```
mov ecx, str_flower  
mov edx, len_flower  
call print_string_len
```

```
mov ecx, border_bot  
mov edx, l_bbot  
call print_string_len
```

```
; 2. Inputs (20 Items)  
mov ESI, p_name  
mov EDI, buf_name  
call get_input
```

```
mov ESI, p_nick  
mov EDI, buf_nick  
call get_input
```

```
mov ESI, p_gender  
mov EDI, buf_gender  
call get_input
```

```
mov ESI, p_age  
mov EDI, buf_age  
call get_input
```

```
mov ESI, p_bday  
mov EDI, buf_bday  
call get_input
```

```
mov ESI, p_hair  
mov EDI, buf_hair  
call get_input
```

```
mov ESI, p_height  
mov EDI, buf_height  
call get_input
```

```
mov ESI, p_town  
mov EDI, buf_town  
call get_input
```

```
mov ESI, p_prog  
mov EDI, buf_prog
```



call get_input

mov ESI, p_work
mov EDI, buf_work
call get_input

mov ESI, p_lang
mov EDI, buf_lang
call get_input

mov ESI, p_happy
mov EDI, buf_happy
call get_input

mov ESI, p_color
mov EDI, buf_color
call get_input

mov ESI, p_movie
mov EDI, buf_movie
call get_input

mov ESI, p_music
mov EDI, buf_music
call get_input

mov ESI, p_singer
mov EDI, buf_singer
call get_input

mov ESI, p_food
mov EDI, buf_food
call get_input

mov ESI, p_hobby
mov EDI, buf_hobby
call get_input

mov ESI, p_book
mov EDI, buf_book
call get_input

mov ESI, p_motto



```
mov EDI, buf_motto  
call get_input
```

```
; 3. Loading Screen  
call clear_screen
```

```
mov ecx, str_load_hdr  
mov edx, len_load_hdr  
call print_string_len
```

```
mov ecx, emo_kirby1  
mov edx, len_emo1  
call print_string_len  
call sleep_delay
```

```
mov ecx, emo_kirby2  
mov edx, len_emo2  
call print_string_len  
call sleep_delay
```

```
mov ecx, emo_kirby3  
mov edx, len_emo3  
call print_string_len  
call sleep_delay
```

```
mov ecx, pbar_0  
mov edx, len_p0  
call print_string_len  
call sleep_delay
```

```
mov ecx, pbar_10  
mov edx, len_p10  
call print_string_len  
call sleep_delay
```

```
mov ecx, pbar_30  
mov edx, len_p30  
call print_string_len  
call sleep_delay
```

```
mov ecx, pbar_60  
mov edx, len_p60  
call print_string_len
```



call sleep_delay

mov ecx, pbar_80
mov edx, len_p80
call print_string_len
call sleep_delay

mov ecx, pbar_100
mov edx, len_p100
call print_string_len
call sleep_delay

mov ecx, str_uploaded
mov edx, len_uploaded
call print_string_len
call sleep_delay

; 4. Final Output Display
call clear_screen

mov ecx, str_out_hdr
mov edx, len_out_hdr
call print_string_len

mov ecx, border_top
mov edx, l_btop
call print_string_len

mov ecx, str_flower
mov edx, len_flower
call print_string_len

mov ecx, border_mid
mov edx, l_bmid
call print_string_len

; Display 20 Fields
mov ESI, lbl_name
mov EDI, buf_name
call print_field

mov ESI, lbl_nick
mov EDI, buf_nick



call print_field

mov ESI, lbl_gender
mov EDI, buf_gender
call print_field

mov ESI, lbl_age
mov EDI, buf_age
call print_field

mov ESI, lbl_bday
mov EDI, buf_bday
call print_field

mov ESI, lbl_hair
mov EDI, buf_hair
call print_field

mov ESI, lbl_height
mov EDI, buf_height
call print_field

mov ESI, lbl_town
mov EDI, buf_town
call print_field

mov ESI, lbl_prog
mov EDI, buf_prog
call print_field

mov ESI, lbl_work
mov EDI, buf_work
call print_field

mov ESI, lbl_lang
mov EDI, buf_lang
call print_field

mov ESI, lbl_happy
mov EDI, buf_happy
call print_field

mov ESI, lbl_color



```
mov EDI, buf_color  
call print_field
```

```
mov ESI, lbl_movie  
mov EDI, buf_movie  
call print_field
```

```
mov ESI, lbl_music  
mov EDI, buf_music  
call print_field
```

```
mov ESI, lbl_singer  
mov EDI, buf_singer  
call print_field
```

```
mov ESI, lbl_food  
mov EDI, buf_food  
call print_field
```

```
mov ESI, lbl_hobby  
mov EDI, buf_hobby  
call print_field
```

```
mov ESI, lbl_book  
mov EDI, buf_book  
call print_field
```

```
mov ESI, lbl_motto  
mov EDI, buf_motto  
call print_field
```

```
mov ecx, border_bot  
mov edx, l_bbot  
call print_string_len
```

```
; Exit Program (`sys_exit`)  
mov eax, 1  
xor ebx, ebx  
int 0x80
```



;

=====

==

; PROCEDURES & HELPER FUNCTIONS

;

=====

==

get_input:

push eax

push ebx

push ecx

push edx

mov ecx, ESI

call print_string_null

mov ecx, EDI

mov edx, 64

call read_string

pop edx

pop ecx

pop ebx

pop eax

ret

print_field:

push eax

push ebx

push ecx

push edx

mov ecx, side_border

mov edx, 1_side

call print_string_len

mov ecx, ESI

call print_string_null

mov ecx, EDI

call print_string_null



```
pop edx
pop ecx
pop ebx
pop eax
ret
```

```
print_string_len:
    push eax
    push ebx
    mov eax, 4      ; sys_write
    mov ebx, 1      ; stdout
    int 0x80
    pop ebx
    pop eax
    ret
```

```
print_string_null:
    push eax
    push ebx
    push ecx
    push edx
```

```
    mov edx, ecx
.count_loop:
    cmp byte [edx], 0
    je .found_end
    inc edx
    jmp .count_loop
.found_end:
    sub edx, ecx    ; Calculate string length
```

```
    mov eax, 4      ; sys_write
    mov ebx, 1      ; stdout
    int 0x80
```

```
pop edx
pop ecx
pop ebx
pop eax
ret
```

```
clear_screen:
    push eax
```



```
push ebx
push ecx
push edx
mov eax, 4
mov ebx, 1
mov ecx, clr_scr
mov edx, clr_len
int 0x80
pop edx
pop ecx
pop ebx
pop eax
ret
```

```
read_string:
push eax
push ebx
mov eax, 3      ; sys_read
mov ebx, 0      ; stdin
int 0x80
pop ebx
pop eax
ret
```

```
sleep_delay:
push eax
push ebx
mov eax, 162    ; sys_nanosleep
mov ebx, timespec
xor ecx, ecx
int 0x80
pop ebx
pop eax
ret
```

OUTPUT:



```
mp2_nasm5.asm umbrella.asm pusheencat.asm asci_art.asm midterm_asci.asm umbrella Ex1TC2.asm Ex1TC3.asm
https://congenial-halibut-7vjrnw4xfjw3rpg4.github.dev
codespaces-blank [Codespaces: CS0003L]
mp2_nasm5.asm umbrella.asm pusheencat.asm asci_art.asm midterm_asci.asm umbrella Ex1TC2.asm Ex1TC3.asm
AsciArt > midterm_asci.asm
1 section_data
Problems Output Debug Console Terminal Ports
@asciervo → /workspaces/codespaces-blank $ nasm -f elf32 midterm_asci.asm -o midterm_asci.o
nasm: fatal: unable to open input file 'midterm_asci.asm' No such file or directory
@asciervo → /workspaces/codespaces-blank/AsciArt $ cd AsciArt
@asciervo → /workspaces/codespaces-blank/AsciArt $ nasm -f elf32 midterm_asci.asm -o midterm_asci.o
@asciervo → /workspaces/codespaces-blank/AsciArt $ ld -m elf_i386 midterm_asci.o -o midterm_asci
@asciervo → /workspaces/codespaces-blank/AsciArt $ ./midterm_asci

SLAMBOOK

/\^ (\^/\)
(o.o) (^x^ ) | [o.o] OFFICIAL ASSEMBLY SLAMBOOK |
> ^ < c(\^
[01] Full Name : Althea Ciervo
[02] Nickname : Thea
[03] Gender : F
[04] Age : 21
[05] Birthday : April 10
[06] Hair Color : Black
[07] Height (m) : 1.68
[08] Hometown : Quezon City
[09] Program & School : BSCSE - FEU Tech
[10] Job/Workplace : N/A
[11] Languages Spoken : Tagalog, English
[12] Happiest Moment : When I'm with my friends

[!] Processing Slambook Memory Buffers... Please Wait!
c('c') (>^>) <('c') ^('c')> (>^>) Q('c') <(^>^>=b
[100%]
(data) Upload complete! Formatting Slambook Card...
```



The screenshot shows a VS Code editor window with a file explorer on the left and a terminal on the right. The file explorer shows a project named 'Midterm Exam/Project' with a file named 'midterm_ascii.asm'. The terminal shows the output of the program, which is an ASCII art representation of a student's information. The output is as follows:

```
1 section .data
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